Statistical Analysis Report

UCI Bank Marketing Dataset — Association Between Job Category and Term-Deposit Subscription

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Overview

This report analyses the UCI Bank Marketing dataset (Moro, Cortez, & Rita, 2014) to examine whether a client's *job category* is associated with whether they subscribe to a term deposit during a direct-marketing telephone campaign by a Portuguese banking institution. The analysis comprises descriptive statistics, four visualisations, and a chi-square test of independence between `job` and the binary subscription outcome `y`. Dataset source: UCI Machine Learning Repository, ID 222 (https://archive.ics.uci.edu/dataset/222/bank+marketing).

Dataset Description

The dataset contains 41,188 rows and 21 columns; each row records one telephone contact with a bank client. Variables include client demographics (`age`, `job`, `marital`, `education`), loan/credit indicators (`default`, `housing`, `loan`), contact information (`contact`, `month`, `day_of_week`, `duration`, `campaign`), prior-campaign history (`pdays`, `previous`, `poutcome`), five macro-economic context variables (`emp.var.rate`, `cons.price.idx`, `cons.conf.idx`, `euribor3m`, `nr.employed`), and the binary target `y` (subscribed: yes / no). The key variables for this report are the categorical predictor `job` (12 categories including *unknown*) and the categorical outcome `y`. The target is highly imbalanced (≈11.3% "yes" / 88.7% "no"), and several categorical columns use the sentinel value *unknown* in place of missing data — `default` (8,597 rows), `education` (1,731), `housing` and `loan` (990 each), `job` (330), `marital` (80).

Methods

Descriptive statistics. I computed the standard `pandas.DataFrame.describe` summary for the eight numeric variables, frequency counts for every categorical variable, the overall and per-job subscription rates, and skewness for the four numeric variables most relevant to client-level inference (`age`, `duration`, `campaign`, `euribor3m`). Initial data analysis of this kind — distribution shape, sentinel/missing patterns, and group-level rates examined *before* any inferential test — is recognised as a foundation of reproducible analysis (Huebner et al., 2018).

Visualisations. Four figures were chosen, each to expose a property that bears on the hypothesis test:

1. *Figure 1.* Histogram of `age`, to characterise the shape of the principal demographic variable.
2. *Figure 2.* Horizontal bar chart of subscription rate by `job`, with the overall rate as a reference line — this exposes the group-level signal that motivates the test.

3. *Figure 3.* Boxplot of `age` by outcome `y`, to compare the two outcome groups.
4. *Figure 4.* Pearson correlation heatmap of the numeric variables, to flag the strong collinearity within the macro-economic block.

Each figure has a descriptive title and labelled axes. Clear axis labels and informative titles are widely recommended in the visualisation literature as a basic quality requirement (Tufte, 2001).

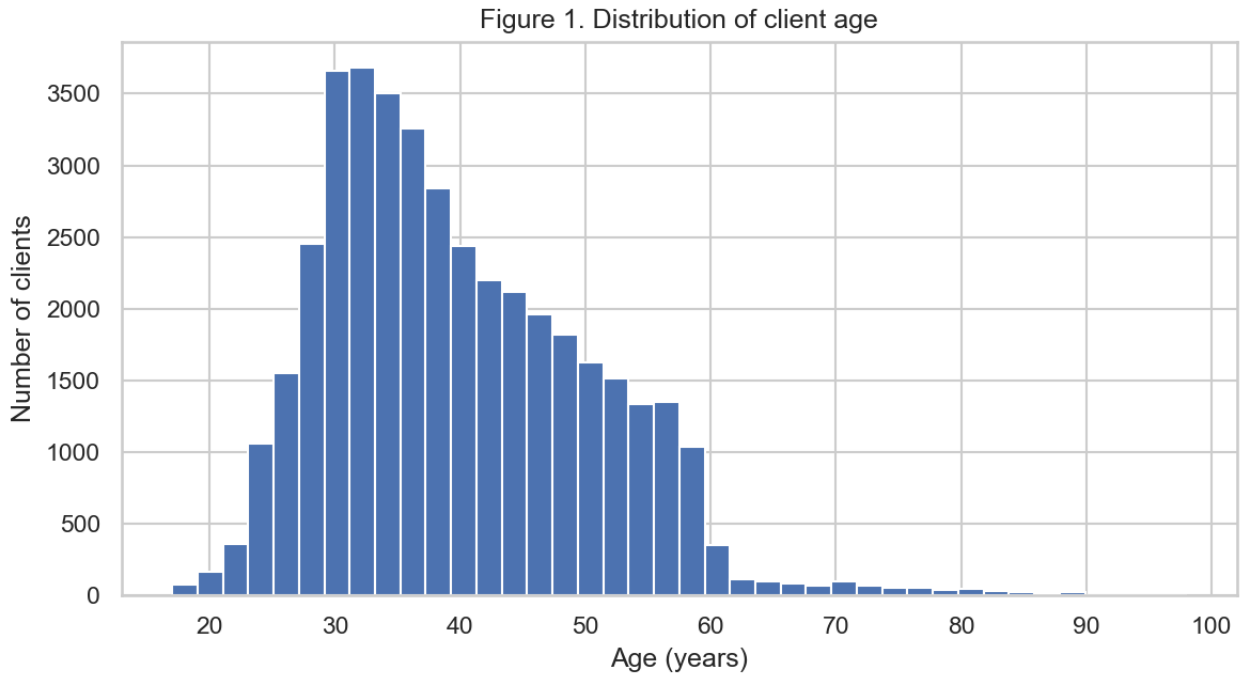

Figure 1. Distribution of client age.

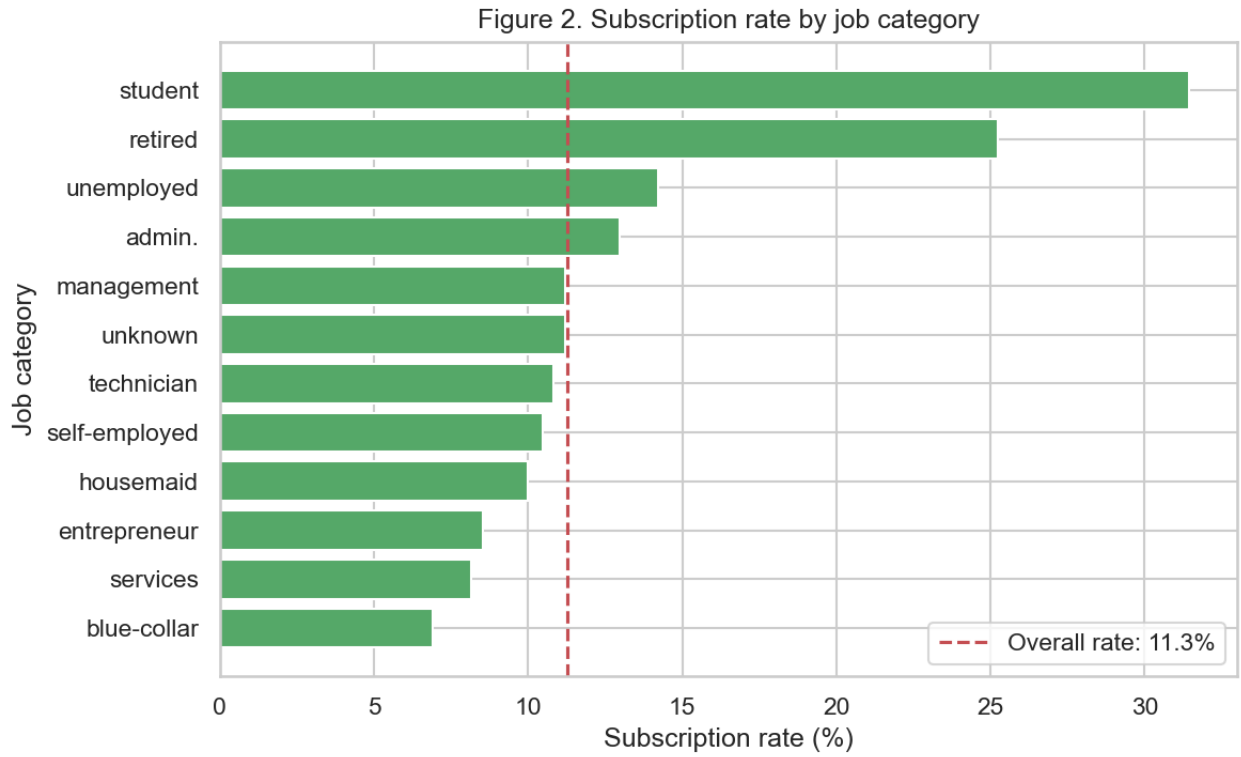

Figure 2. Subscription rate by job category, with the overall rate as a reference line.

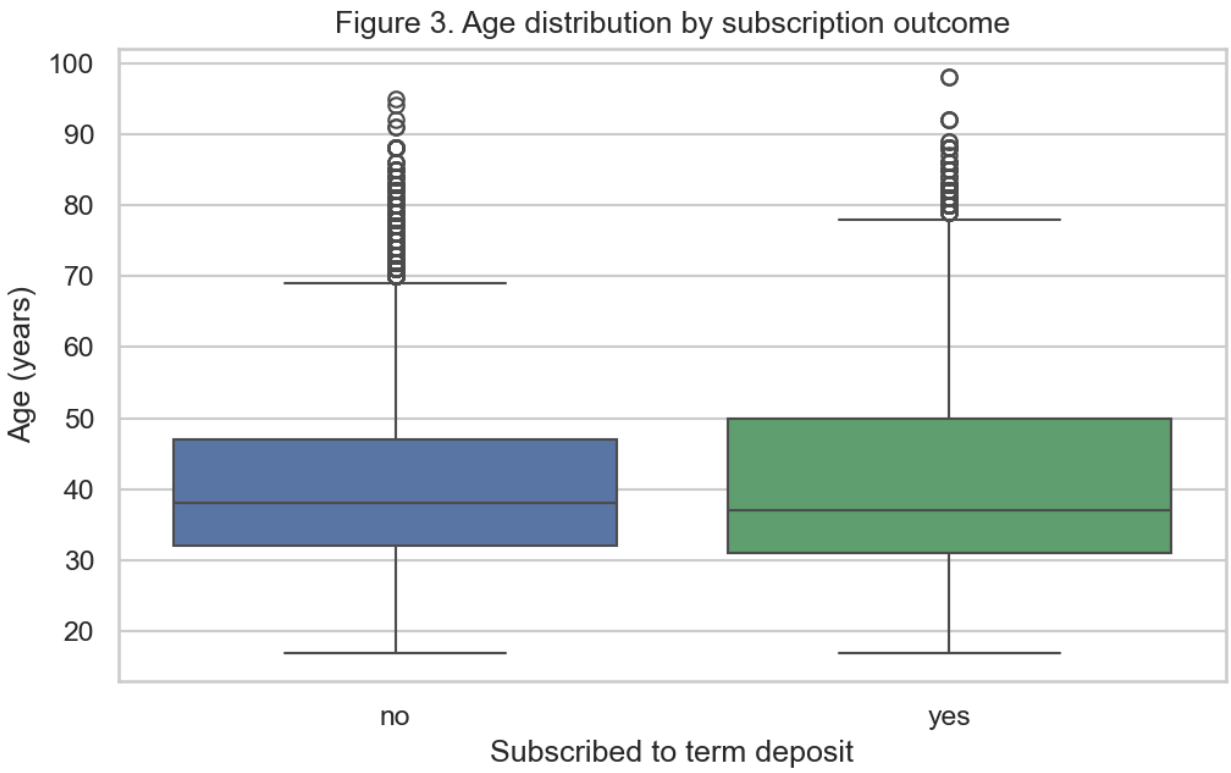

Figure 3. Age distribution by subscription outcome.

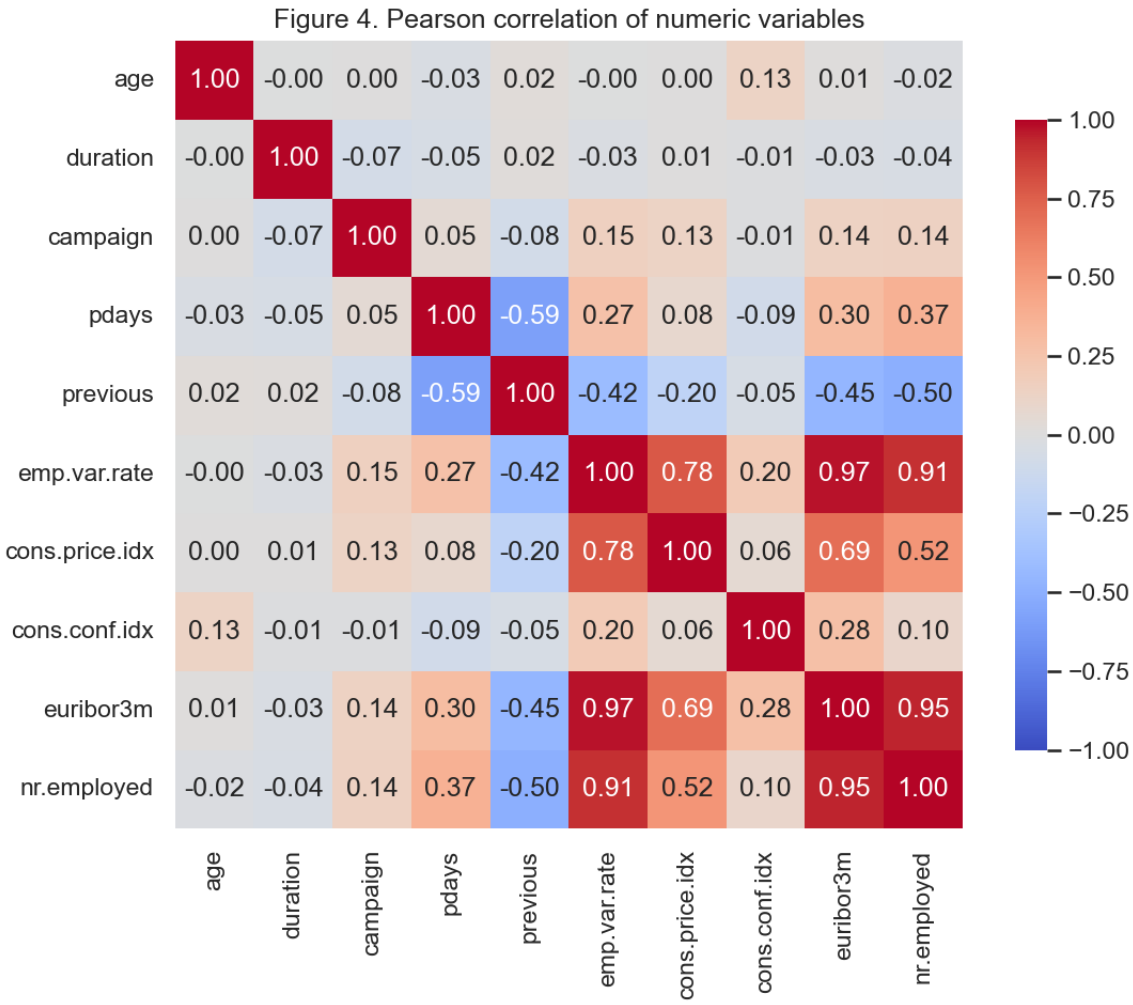

Figure 4. Pearson correlation heatmap of numeric variables.

Hypothesis test. I tested whether subscription outcome is independent of job category using **Pearson's chi-square test of independence** (Agresti, 2007) on the 12 × 2 contingency table of `job` × `y`.

- H_0: subscription outcome is independent of job category.
- H_1: subscription outcome is associated with job category.
- Significance level: $\alpha = 0.05$.

The chi-square test is appropriate here because (i) both variables are categorical, (ii) the 41,188 observations are independent (one row per client contact), and (iii) the assumption that all expected cell counts are at least 5 is satisfied — the minimum expected count in our table is 37.2 (Agresti, 2007). To gauge the practical magnitude of any association, I also report Cramer's V (Cohen, 1988), interpreted on the standard small / medium / large scale (≈0.10 / 0.30 / 0.50) for an effect with min(rows, cols) - 1 = 1 degree of freedom.

Results

Descriptive findings. The overall subscription rate is 11.3%. Subscription rates by job range from 6.9% (*blue-collar*) and 8.1% (*services*) at the low end to 25.2% (*retired*) and 31.4% (*student*) at the high end (Figure 2). Numeric variables show heavy right-skew for `duration` (skew = 3.26) and `campaign` (skew = 4.76), mild right-skew for `age` (0.78), and bimodality for `euribor3m`. The macro-economic block is strongly intercorrelated, with |r| > 0.75 for several pairs (Figure 4).

Hypothesis test. Pearson's chi-square test of independence gave $\chi^2 = 961.24$ on dof = 11, $p \approx 4.2 \times 10^{-199}$. The minimum expected cell count was 37.2, well above the conventional threshold of 5 (Agresti, 2007), so the asymptotic chi-square approximation is appropriate. Cramer's V was 0.153, a small-to-moderate effect (Cohen, 1988). At $\alpha = 0.05$, we reject H_0 and conclude that subscription outcome is associated with job category.

Interpretation for a Non-Technical Audience

Across roughly 41,000 phone calls, about 11 in every 100 clients agreed to open a term deposit. That overall rate hides large differences across professions. Students said yes about 31% of the time, retirees about 25%, while blue-collar and services workers said yes only 7–8% of the time — well under half the overall rate. A statistical test confirmed that this difference is far too large to be a coincidence of the sample: the chance of seeing a pattern this strong in random data, if profession truly made no difference, is essentially zero. That said, knowing a client's profession only modestly improves your ability to predict the outcome — many other things matter too. The practical takeaway is that profession is a real and statistically reliable signal, but it is not by itself a strong predictor.

Limitations and Potential Bias

Limitation — data was collected during a single economic period. The five macro-economic variables (`emp.var.rate`, `euribor3m`, `nr.employed`, `cons.price.idx`, `cons.conf.idx`) vary over only a narrow range and are highly correlated with each other (Figure 4). Subscription behaviour during this window may not generalise to a period with very different interest-rate or employment conditions. This is the classic *temporal-context limitation* of single-window observational data (Huebner et al., 2018).

Bias — selection and channel bias. The dataset records only clients that the bank *chose to call* through a particular outbound-telemarketing programme. Clients who do not own a phone, do not answer unknown numbers, or were excluded from the institution's call list at the outset are systematically absent. Subscription rates measured here therefore reflect the population *reachable and willing to take a marketing call*, not the bank's overall customer base. Any generalisation to "all bank clients" would be biased upward for groups with higher call-pickup rates (e.g. *retired* clients, who are home more often) and downward for groups less likely to answer cold calls.

Additional caveat. The variable `duration` (call length in seconds) leaks the outcome — it is recorded only after the call ends, and a non-subscription is sometimes a very short call. Per the dataset's own documentation, `duration` should not be used as a predictor of `y`; I used it for descriptive context only.

References

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